MIL-G-47316(MI)
14 August 1987

MILITARY SPECIFICATION

GRAPHITE, MOLDED, FOR HIGH TEMPERATURE APPLICATIONS

This specification is approved for use within by the Army Missile Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements of one type of molded graphite for high temperature applications.

1.2 Classification. The molded graphite shall be of the following styles:

STYLES

Style a - rectangular blocks
Style b - cylindrical billets

1.3 Part number. The military part number shall consist of the letter B, the basic number of this specification and dash numbers denoting style (see table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 9390

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Example:

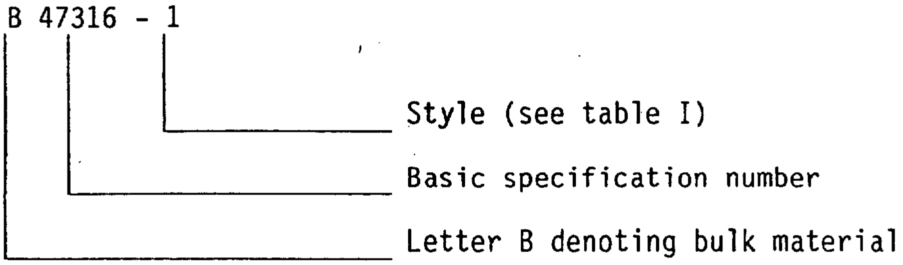


TABLE I. Style identification.

Style	Dash identification number
a--------------	-1
b--------------	-2

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-794	Parts and Equipment, Procedures for Packaging of

(Copies of standards required by contractors in connection with acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C78	Flexural Strength of Concrete (Using Simple Beam with Third-Point Loading), Standard Test Method for

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational sources.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Material. The material shall be graphite formed by the molding process from petroleum coke having a particle size not greater than 0.006 inch combined with coal tar pitch and shall conform to the requirements of this specification (see 4.6.6).

3.2.1 Dimensions. The dimensions of the graphite rectangular blocks and cylindrical billets shall be as specified in the acquisition documents (see 6.2.1).

3.3 Properties.

3.3.1 Apparent density. Apparent density of the graphite shall be not greater than 1.83 nor less than 1.70 grams per cubic centimeter (g/cc) (see 4.6.2).

3.3.2 Specific resistance. Specific resistance of the graphite shall be not greater than 13.0 x 10^{-4} ohm - centimeter (ohm - cm) with the grain and not greater than 17.0 x 10^{-4} ohm - cm across the grain (see 4.6.3).

3.3.3 Flexural strength. Flexural strength of the graphite shall be not less than 2600 pounds per square inch (psi) with the grain and not less than 2400 psi across the grain (see 4.6.4).

3.3.4 Ash content. Ash content of the graphite shall be not greater than 0.25 percent (see 4.6.5).

3.4 Marking. Each block or billet of graphite defined by this specification shall be permanently marked in accordance with MIL-STD-130 with the following information:

a. Title, number and date of this specification.

b. Part number.

c. Supplier's name, address, and material designation.

d. Purchase order number.

e. Lot number.

f. Weight and size of block or billet.

3.4.1 Identification. Rectangular blocks and cylindrical billets shall be color-coded red on each end. The with-grain direction shall be marked with an arrow in accordance with figure 1.

3.5 Workmanship. The graphite shall be manufactured to assure uniform material free from visual cracks or other defects that would prevent its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

   (1) One hundred percent inspection.

   (2) Sampling inspection.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the conditions of 4.3.1 and 4.3.2 of this specification.

4.3.1 Ambient conditions. Unless otherwise specified, all inspections and tests shall be conducted at room temperature between 20.9 and 31.9° Celsius (C) [70 and 90° Fahrenheit (F)] and at atmospheric pressure.

4.3.2 Test specimens. From the sample block or billet, a minimum of three test specimens shall be obtained across-the-grain and a minimum of three test specimens shall be obtained with-the-grain as shown in figure 1. Dimensions of the test specimens shall be specified in the acquisitions documents (see 6.2.1).

4.4 First article inspection. First article inspection shall consist of the examination and tests as specified under 4.6, and shall be conducted in the order specified in 4.4.1. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity (see 3.1 and 6.2).

4.4.1 First article sample. The first article sample shall consist of ten blocks or billets, as specified in the acquisition documents, representative of the production product. The first article samples shall be subjected to inspection and tests specified in table II in the order shown.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Examination of product------	3.4, 3.4.1 & 3.5	4.6.1
Apparent density-----------	3.3.1	4.6.2
Specific resistance---------	3.3.2	4.6.3
Flexural strength----------	3.3.3	4.6.4
Ash content-----------------	3.3.4	4.6.5
Certification---------------	3.2	4.6.6

4.5 Quality conformance inspection.

4.5.1 One hundred percent inspection. Each block or billet to be delivered shall be subjected to the inspection of table III in the order shown.

TABLE III. One hundred percent inspection.

Inspection	Requirement paragraph	Test method paragraph
Examination of product------	3.4, 3.4.1 & 3.5	4.6.1
Apparent density-----------	3.3.1	4.6.2

4.5.2 Sampling inspection by lot. Unless otherwise specified by the acquisition documents, a sample selected from each lot of blocks or billets submitted for acceptance shall be subjected to inspections in addition to the one hundred percent inspection to insure that quality of product is maintained during production.

4.5.2.1 Inspection lot. A lot shall consist of graphite of one shape produced without change in material or process and offered for acceptance at one time. The lot size shall not exceed 100 blocks or billets.

4.5.2.2 Sampling. A sample shall consist of a random selection of 10 percent of the lot but not less than one block or billet, as specified by the acquisition documents.

4.5.2.3 Sample inspection. Samples selected in accordance with 4.5.2.2, in addition to one hundred percent inspection, shall be subjected to the inspections of table IV in the order shown.

TABLE IV. Sample inspection.

Inspection	Requirement paragraph	Test method paragraph
Specific resistance------	3.3.2	4.6.3
Flexural strength--------	3.3.3	4.6.4
Ash content--------------	3.3.4	4.6.5

4.5.3 Noncompliance. Failure of any sample of block or billet to conform to apparent density, identification and workmanship shall be cause for rejection of the individual block. Failure of the average test results of any sample of block or billet to conform to specific resistance, flexural strength or ash content shall be cause for rejection of the lot represented by the sample.

4.5.4 Inspection of packaging. The preservation, packaging, packing and marking shall be inspected for conformance to section 5.

4.6 Methods of inspection.

4.6.1 Visual examination. Samples shall be visually examined to determine compliance with 3.4, 3.4.1 and 3.5. Unless otherwise specified, all visual examinations shall be conducted with an unaided eye.

4.6.2 Apparent density. Apparent density (see 3.3.1) shall be determined in accordance with the following:

a. Measure each dimension of the full block or billet to the nearest 0.03 inch in three or more places.

b. Average each dimension and calculate the volume of the block or billet based on the average dimensions.

c. Weigh the block or billet to the nearest 0.25 pound.

d. Calculate apparent density as follows:

$$\text{Apparent density, g/cc} = \frac{M(27.68)}{V}$$

Where: M = weight of sample, pounds

V = volume of sample, cubic inches

27.68 = conversion factor to grams per cubic centimeter

e. Report the apparent density of each block or billet in the lot.

4.6.3 Specific resistance. Specific resistance (see 3.3.2) shall be determined in accordance with the following:

a. Assemble specific resistance testing apparatus as shown in figure 2.

b. Balance the potentiometer according to the manufacturer's instructions.

c. Use specimens A (see figure 1) for with-the-grain determinations and specimens B for across-the-grain determinations.

d. Place a weight of approximately 5 pounds on the center of the specimen to assure contact of the potential points.

e. Determine the current passing through the test specimen.

f. Determine the voltage drop across the potential points.

g. Calculate specific resistance as follows:

$$\text{Specific resistance, } 10^{-4} \text{ ohm-cm} = \frac{E \times A}{I \times L}$$

Where: A = cross-sectional area, square cm

L = length of conductor, cm

E = potential difference, volts

I = current, amperes

h. Report the average of the results for the A specimens (with grain) and the average for the B specimens (across grain) for each block or billet.

4.6.4 Flexural stength. Flexural strength (see 3.3.3) shall be determined in accordance with ASTM C78 and the following:

a. Optional construction of the loading head and bed of the testing machine shall be permissible provided the distances between supports and points of load application are maintained and the application of load and directions of reactions are not affected.

b. The contact surface of each load applying and load receiving block shall be cylindrical or have a 0.03 inch wide flat surface. (No sharp edges are permitted.)

c. Only specimens that fracture within the middle third of the span length shall be considered valid tests.

d. Use specimens A for with-the-grain determinations and specimens B for across-the-grain determinations.

e. Report the average of the results with-the-grain and the average across-the-grain for each sample block or billet.

4.6.5 Ash content. Ash content (see 3.3.4) shall be determined in accordance with the following using composited pieces of test specimens from the flexural strength test:

a. Weigh a sample of approximately 10 g, to the nearest milligram (mg), into a tared 100 cubic centimeter (cc) platinum or porcelain crucible.

b. Place the weighed sample in a muffle furnace and heat the sample to a temperature of 800 to 900° C(1497.6 to 1677.6°F) for a minimum of 24 hours. To prevent ash fusion, do not exceed 900°C(1677.6°F).

c. Allow the crucible and sample to cool to room temperature in a desiccator.

d. Examine the residue for traces of carbon. If carbon is present, continue ignition until ashing is complete.

e. Transfer the sample residue to a desiccator and allow to stand for a minimum of 1 hour before weighing.

f. Weigh the residue to the nearest 0.1 mg.

g. When there is no further visual evidence of carbon, repeat the ignition for 1 hour periods at 800 to 900° C(1497.6 to 1677.6°F) until a constant weight is obtained.

h. Calculate ash content as follows:

$$\text{Ash content, percent} = \frac{\text{ash weight}}{\text{sample weight}} \times 100$$

i. Report the average of a minimum of two determinations for each sample block or billet.

4.6.6 Certification. The supplier shall certify that the material meets the requirements of 3.2.

5. PACKAGING

5.1 Preservation. Preservation shall be in accordance with level C of MIL-STD-794.

5.2 Packaging. Packaging shall be in accordance with level C of MIL-STD-794.

5.3 Packing. Packing shall be in accordance with level C of MIL-STD-794.

5.4 Marking. Marking of each shipping container shall be in accordance with MIL-STD-129, and shall include, but not be limited to, the following:

a. Title, number and date of this specification.

b. Part number.

c. Manufacturer's grade designation.

d. Manufacturer name.

e. Lot number.

f. Weight and size of blocks or billets.

g. Contract or purchase order number.

6. NOTES

6.1 Intended use. Graphite in accordance with this specification is intended for use in rocket motor nozzle components.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Part number (see 1.3).

c. Whether first article sample required (see 3.1 and 4.4).

d. Size of product required (see 3.2.1).

e. Size of test specimens required for inspection (see 4.3.2).

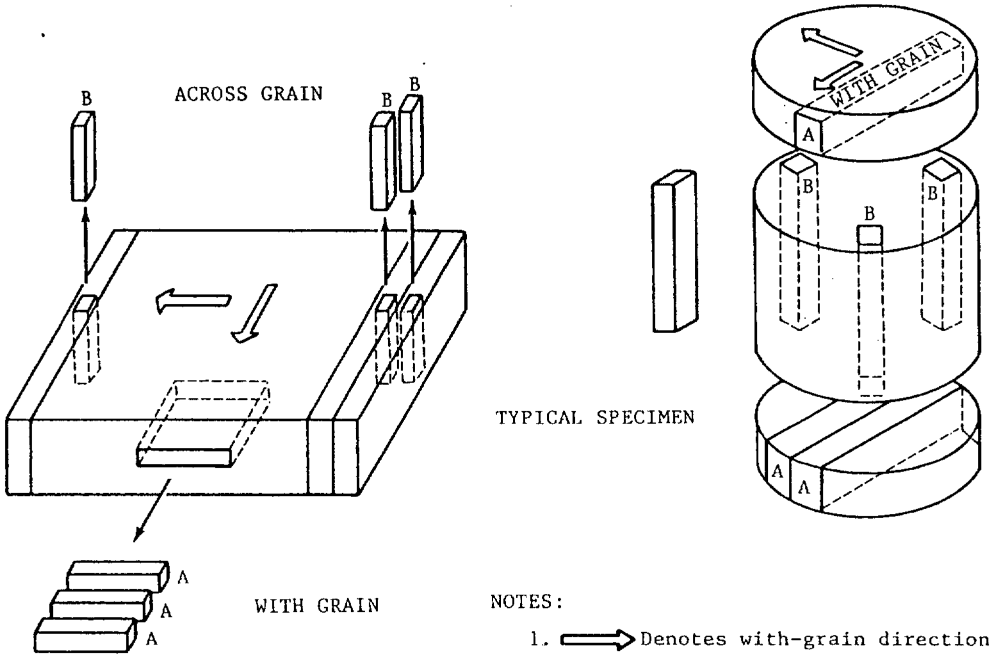
6.2.2 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of ten units. The contracting officer should include specific instructions in acquisition documents regarding arrangement for examination, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Subject term (keyword) listing.

Graphite, high temperature, block
Graphite, high temperature billet

Custodian: Army - MI	Preparing activity: Army - MI
	(Project No. 9390-A001)

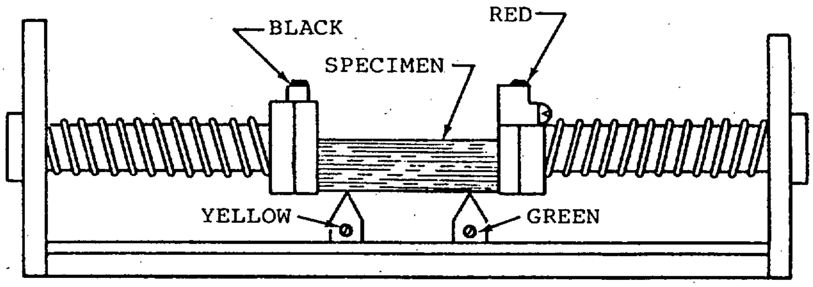
☆ U. S. GOVERNMENT PRINTING OFFICE: 1987—704-036/74465


B
ACROSS GRAIN
B
B
A
A
A
WITH GRAIN
TYPICAL SPECIMEN
WITH GRAIN
A
B
B
B
A
A

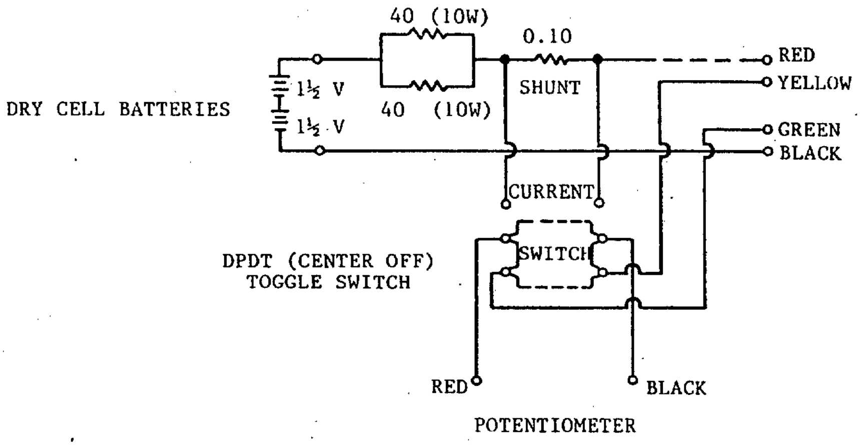
NOTES:

1. ⇨ Denotes with-grain direction
2. Specimen A with grain
3. Specimen B across grain

FIGURE 1. Graphite block and billet test specimens.


BLACK
RED
SPECIMEN
YELLOW
GREEN

RESISTANCE APPARATUS


40 (10W)
0.10
RED
SHUNT
YELLOW
DRY CELL BATTERIES
1½ V
1½ V
40 (10W)
GREEN
BLACK
CURRENT
SWITCH
DPDT (CENTER OFF)
TOGGLE SWITCH
RED
BLACK
POTENTIOMETER

ELECTRICAL SCHEMATIC

FIGURE 2. Specific resistance test arrangement.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER MIL-G-47316(MI)	2. DOCUMENT TITLE GRAPHITE, MOLDED, FOR HIGH TEMPERATURE APPLICATIONS
3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)* ☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: ______
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 82 MAR

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